User Manual: Distributed Math Wizard

Overview

The Distributed Math Wizard simulates a high-performance, decoupled architecture. It is designed to handle heavy background computations without interrupting the experience of casual users browsing the front-end dashboard.

Core Features and Navigation

- **UI Dashboard:** Access the root URL (`/`) to view the main graphical interface.
- **Identity Check:** Navigate to `/whoami` to see the specific container ID of the node currently serving your request. Due to round-robin load balancing, this ID will change as requests are routed to different replicas.
- **System Diagnostics:** Navigate to `/valkey_test` to verify that the application layer is successfully communicating with the shared state database.

Submitting Math Tasks

- **Synchronous Processing:** Use the `/add` endpoint. This will block the specific API container until the math is finished, simulating a legacy architecture.
- **Asynchronous Processing:** Submit a `POST` request to the `/task` endpoint. The system will instantly accept the job, push it to a background Valkey queue, and return a unique `task_id`.

Tracking Task Status

- Use the `/tasks/<id>/status` endpoint, replacing `<id>` with your generated task ID.
- The system will query the shared state and return the status (e.g., PENDING, COMPLETED).
- The response will also detail which specific Worker ID processed the math and which API ID served the status update.

System Safety Limits

- **Polling Timeout:** If you check the status of a task 10 times and it has not completed, the client will abort the request and signal a "Server Busy" state to prevent infinite loops.
- **Circuit Breaker:** If the background queue backs up with more than 20 pending tasks, the system will automatically reject new submissions with a `503 Service Unavailable` error to protect the database from crashing.